

Sec-A

(1) What is Seek Time?

Ans Seek time refers to the time it takes for the read/write head of a disk drive to move from its current location to the desired track on the disk where the data is stored. It is one of the major components of disk access time, which also includes rotational latency and transfer time.

(2) Explain Rotational Latency in Disk scheduling?

Ans Rotational latency is the time it takes for the desired sector of a disk to rotate under the read/write head so that data can be accessed. It is one of the key components of disk access time, which significantly impacts disk performance and overall system responsiveness.

3 write the directory structure.

Ans The directory structure is the organization of files into a hierarchy of folders. It should be stable and scalable; it should not fundamentally change, only be added to. Computers have used the folder metaphor for decades as a way to help users keep track of where something can be found.

These are the common top-level directories associated with the root directory:

Directories	Description
/bin	- binary or executable programs.
/etc	- System configuration files
/usr	- User related programs.
/tmp	- Temporary space

Dec 22, 2023, 17:40 Optional or third party software

/var - log files.

Q

Explain I/O subsystems.

Ans

The I/O (input/output) subsystem is a crucial component of any computer system. It acts as a bridge between the internal processing units and the external environment, allowing users to interact with the computer and exchange data with other devices.

S

Explain the concept of file sharing?

Ans

File sharing is the practice of distributing or providing access to digital information or resources such as documents, multimedia (audio, video, images), graphics, computer programs, and ebooks. The primary purpose of file sharing is to allow users to access, read, write, or execute files and resources that are stored on a different computer or server.

(Sec-B)

(1) Role of I/O buffering in I/O management with suitable diagram.

Ans

I/O buffering plays a crucial role in improving the efficiency and performance of computer systems. It acts as a temporary storage location for data being transferred between the CPU and I/O devices. Essentially, it helps bridge the gap between the fast processing speed of the CPU and the slower speed of most I/O devices.

• Benefits of I/O Buffering:

- (i) Reduced CPU overhead: By buffering data, the CPU doesn't need to wait for the I/O device to complete each individual transfer. This frees up CPU time for other tasks, leading to better overall system performance.

(ii) Improved I/O device performance : Buffering allows the I/O device to transfer data in larger chunks, which is more efficient than transferring data one byte at a time. This can significantly improve the performance of I/O operations.

(iii) Reduced latency : Data can be retrieved from the buffer much faster than it can be retrieved from the original source. This can improve the responsiveness of applications and reduce user-perceived latency.

(iv) Reduced seek time : When buffering disk reads, the OS can prefetch additional data beyond what is initially requested. This can reduce the number of times the disk head needs to move, which can significantly improve performance.

• Types of I/O Buffering :-

(i) Single buffering : Only one buffer is used for each I/O operation. This is the simplest type of buffering, but it can be less efficient than other methods.

(ii) Double buffering : Two buffers are used for each I/O operation. This allows one buffer to be filled while the other buffer is being emptied, which can improve performance.

(iii) Circular buffering : Data is written to and read from a circular buffer. This can be an efficient way to buffer data that is being streamed from one device to another.

I/O buffering is an essential technique for improving the performance and efficiency of I/O operations. By understanding the benefits of I/O buffering the different types of buffering available, system designers can

Choose the appropriate buffering strategies to optimize performance for their specific needs.

2. Write the Issues of File implementation system in OS.

Ans File system implementation in an OS involves various challenges and issues. Here are some common problems associated with file system implementation:

- **Data Integrity:** Ensuring the integrity of data is a critical concern. File systems must prevent data corruption or loss due to system crashes, power failures, or hardware faults.
- **Concurrency Control:** When multiple processes or users access the file system simultaneously, concurrency control becomes crucial to prevent conflicts, such as data corruption.
- **File organization and Efficiency:** Designing an efficient file organization is challenging. File systems need to balance factors like quick access times, minimal storage space utilization, and efficient use of disk blocks.
- **File Naming:** Managing file names and directories efficiently, handling naming conflicts, and ensuring a meaningful naming convention can be challenging, especially in large file systems.
- **Disk Space Management:** Efficiently managing disk space, handling fragmentation, and avoiding both external and internal fragmentation are critical issues in file system implementation.
- **File System Fragmentation:** Over time, file systems can experience fragmentation, leading to scattered

storage blocks. This can impact performance as it may take longer to access fragmented files.

3 Give the impact of security in file system implementation as well as in protection.

Security is a critical aspect of file system implementation and protection is an operating system. The impact of security in these areas is profound and influences the overall integrity, confidentiality, and availability of data. Here are the key impacts of security in file system implementation and protection.

(i) Data integrity:

- Impact: Security measures ensure that data remains intact and unaltered. Protection against unauthorized modifications or corruption is crucial for maintaining the integrity of files and the file system.

(ii) Access control:

Impact: Security in access control ensures that only authorized users or processes can access specific files. Proper authentication and authorization mechanisms prevent unauthorized access and protect against data breaches.

(iii) Protection Against Malware:

- Impact: Security measures help prevent the introduction and spread of malware within the file system. This includes implementing antivirus solutions, monitoring for suspicious activities, and restricting permissions.

(iv) Secure File Transmission:

- Impacts: In distributed systems or networked environments, security measures such as encryption and secure protocols are essential for protecting files during

transmission, Preventing eavesdropping or unauthorized interception.

(v) Data Availability:

- Impacts: Security measures contribute to data availability by preventing unauthorized disruptions or attacks that could lead to data loss or system downtime.

(vi) Data Backups and Recovery:

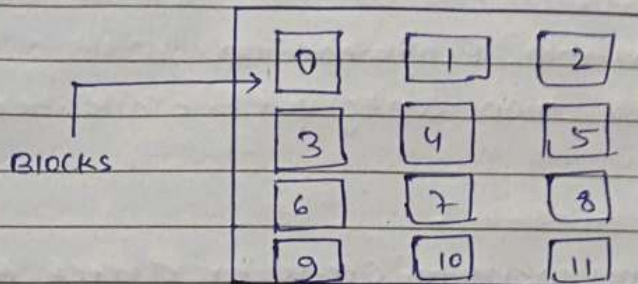
- Impacts: Security considerations include implementing robust backup and recovery mechanisms. In the event of security incidents, having reliable backups ensures that data can be restored, minimizing the impact of data loss or system compromise.

Part - C

(1) Explain disk space allocation method (contiguous, linked, Indexed, I-Node) with suitable diagram.

Ans Disk space allocation methods determine how files are stored on a disk and how the OS keeps track of the locations of these files. There are several disk space allocation methods, each with its advantages and disadvantages. Here's an explanation of four common disk space allocation methods:

(1) Contiguous Allocation: In contiguous allocation, each file occupies a contiguous block of storage on the disk. Files are stored in a single, continuous sequence of disk blocks.

Contiguous AllocationHard Disk

File Name	Start	Length	Allocated Blocks
file1.txt	0	4	0, 1, 2, 3
sun.jpg	5	3	5, 6, 7
mov.mp4	9	3	9, 10, 11

Directory

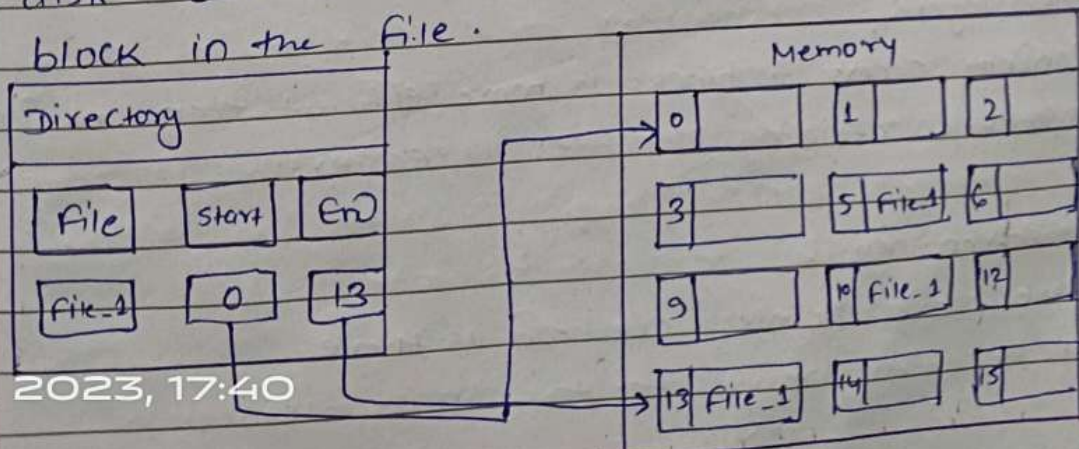
• Advantages of Contiguous Allocation:-

- it is very easy to implement
- There is a minimum amount of seek time.
- Memory access is faster.

• Disadvantages:

- Fragmentation can occur over time, leading to wasted space.
- Dynamic resizing of files is challenging.

(2) Linked Allocation: in linked allocation, each file is a linked list of disk blocks scattered across the disk. Each block contains a pointer to the next block in the file.



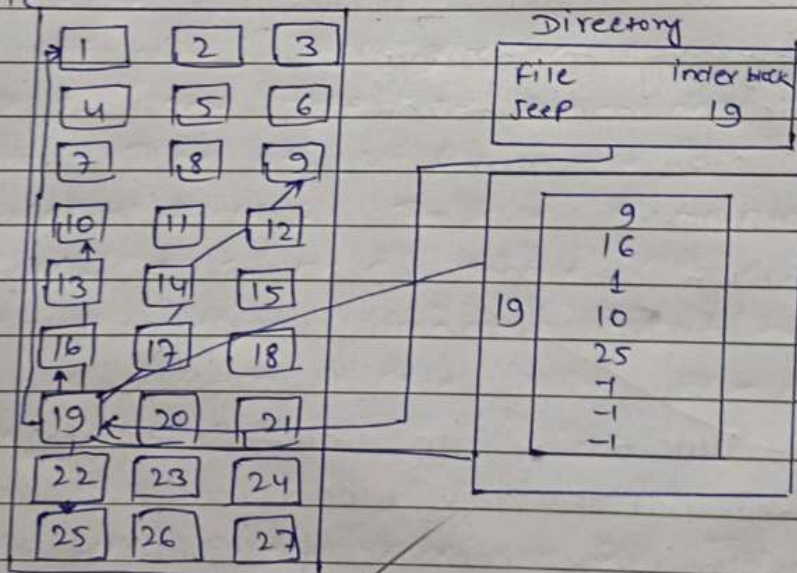
- Advantages:

- There is no external Fragmentation.
- It is more flexible than contiguous file allocation.

- Disadvantages:

- it does not support random access or direct access.
- If pointer are affected so the disk blocks are also affected.

(2) Indexed Allocation: In indexed allocation, a separate index block is used to store pointers to each block of the file. The index block contains the addresses of all the blocks that make up the file.



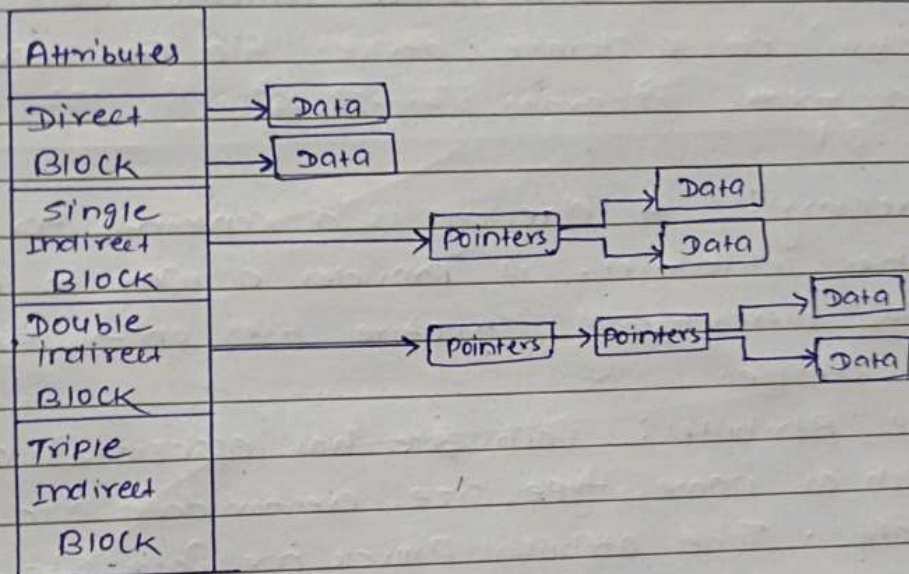
- Advantages:

- Random access to blocks is more efficient than linked allocation.
- Dynamic resizing of files is easier.

- Disadvantages:

- Here more pointer overhead is there.
- it becomes heavy for the small files.

- (4) I-Node (Index Node) Allocation: Commonly used in Unix-like file systems, each file is represented by an index node (i-node). The i-node contains information about the file and pointers to the data blocks that store the file's content.



- Advantages:
 - Efficient random access to both file metadata and data blocks.
 - Supports symbolic and hard links.
- Disadvantages:
 - Extra space is needed for storing i-nodes.
 - Limited scalability for very large file systems.

Q 2

Explain the concept of file system in I/O management.

Discuss its structure and access mechanism.

Ans

A file system is a crucial component of I/O (input/output) management in an OS. It provides a structured and organized way to store, retrieve, and manage files on storage devices such as hard disk, solid state drives and other storage media. The

File System abstracts the physical storage devices, presenting users and applications with a logical hierarchy of files and directories.

- File system structures:

- (i) File: A file is a named collection of data that is stored on a storage device. Files can represent documents, programs, images, etc.
- (ii) Directory: A directory is a container for files and other directories. It provides a way to organize and hierarchically structure files on the storage device.
- (iii) File Attributes: Each file has associated attributes such as name, type, size, creation date, modification date. These attributes provide metadata about the file.
- (iv) File Control Block: The File Control Block (FCB) or Inode is a data structure associated with each file. It contains information about the file, such as the location of data blocks, permissions, and timestamps.

- File System Access Mechanisms: The file provides different mechanisms for accessing and manipulating data.

- (i) Open: Creates a connection between the user program and the file, enabling reading, writing, or both.
- (ii) Read: Retrieves data from the file.
- (iii) Write: Stores data into the file.
- (iv) Delete: Removes a file from the file system.

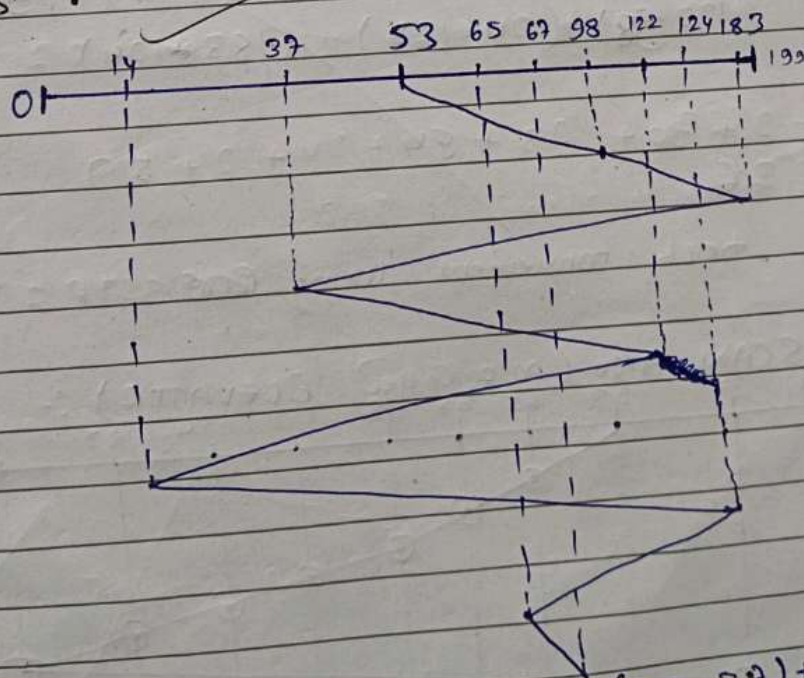
• Different Access Mechanism

- (i) Direct access: Accesses a specific block directly by its address, offering the fastest access but requiring knowledge of the block location.
- (ii) Random access: Allows reading or writing data from any location within the file; ideal for smaller files or when access patterns are unpredictable.
- (iii) Sequential access: Reads or writes data one block at a time in a sequential order, suitable for large files where access patterns are predictable.

Q3. in disk scheduling, find the total track movement by R/W heads using FCFS, SSTF, SCAN, C-SCAN, LOOK C-LOOK. The given track request is 98, 183, 37, 122, 14, 124, 65, 67. The current position of R/W head is at 53.

Ans

FCFS :



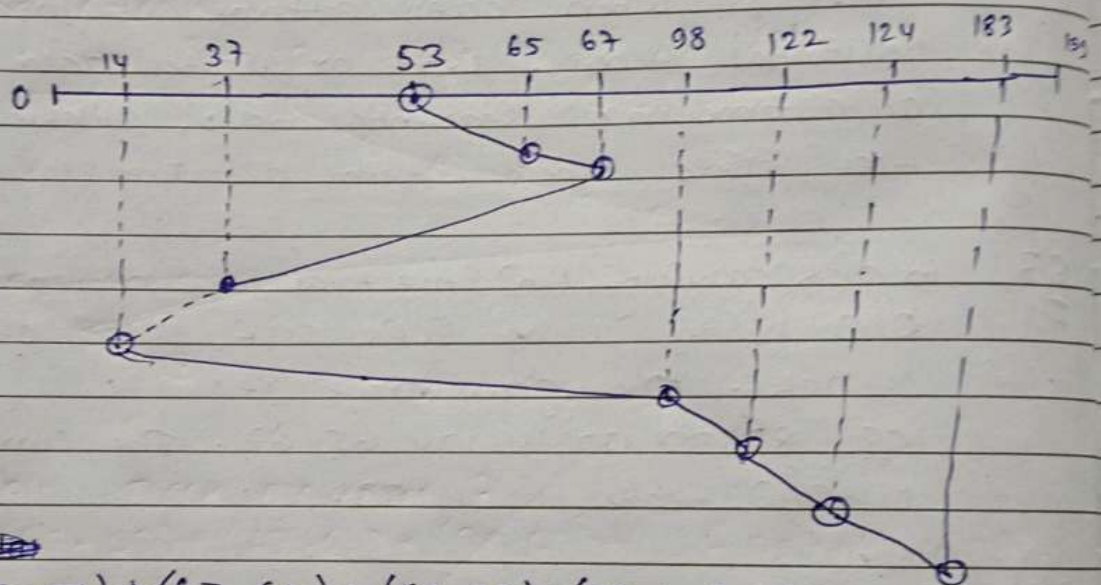
$$= (98-53) + (183-98) + (183-37) + (122-37) + (122-14) + (124-14) + (124-65) + (67-65) =$$

$$= 45 + 85 + 146 + 85 + 108 + 110 + 59 + 2$$

$$= 640$$

Total Movement using FCFS = 640

• Wing SSTF :



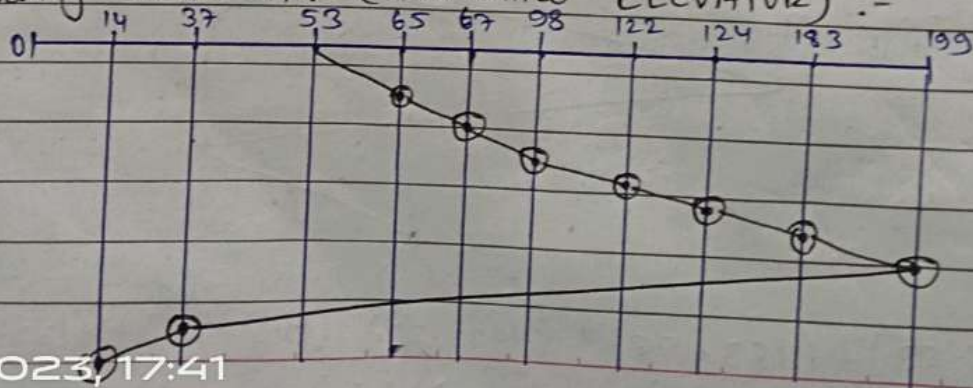
$$= (65-53) + (67-65) + (67-37) + (37-14) + (98-14) + (122-98) + (124-122) + (183-124) =$$

$$= 12 + 2 + 30 + 23 + 84 + 24 + 2 + 59$$

$$= 236$$

Total track movement using ~~FCFS~~ SSTF = 236

• Using SCAN Algo (Also called ELEVATOR) :-

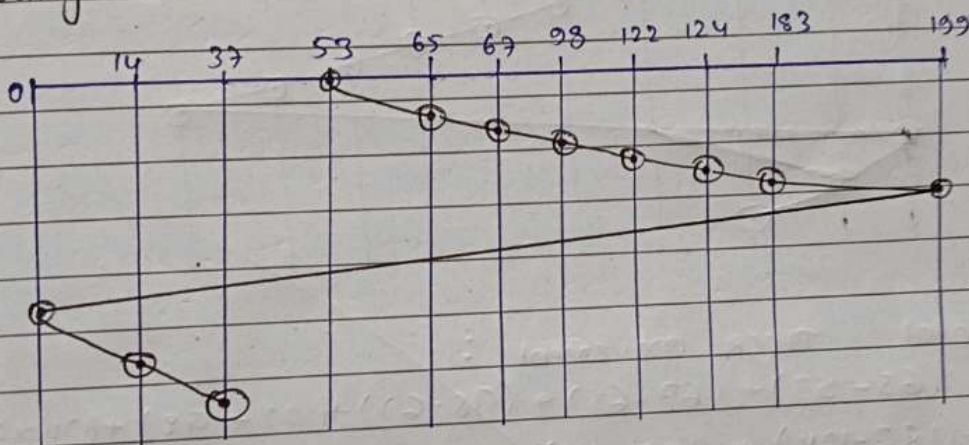


Total Track Movement =

$$= (65-53) + (67-65) + (98-67) + (122-98) + (124-122) + (183-124) \\ + (199-183) + (199-37) + (37-14)$$

$$\Rightarrow 12 + 2 + 31 + 24 + 2 + 59 + 16 + 162 + 23 \\ = 331 \text{ m}$$

• Using C-SCAN (Circular SCAN) :-

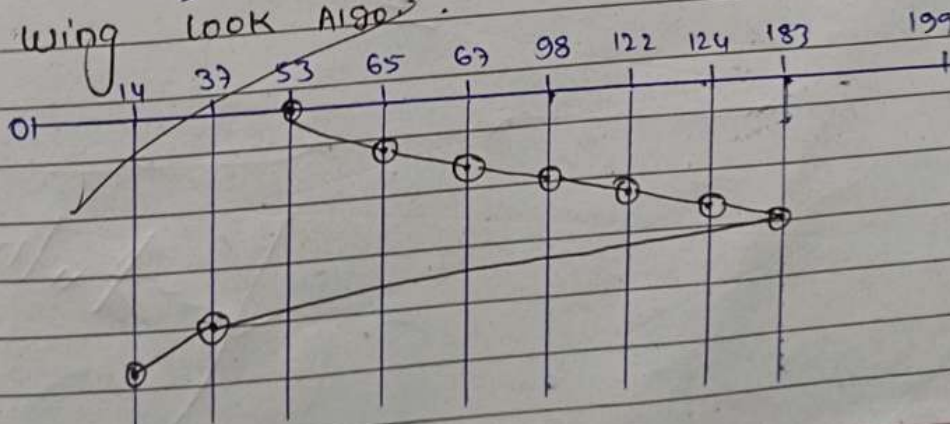


$$\text{Total Track Movement} = \\ = (65-53) + (67-65) + (98-67) + (122-98) + (124-122) + (183-124) \\ + (199-183) + (199-0) + (14-0) + (37-14)$$

$$\Rightarrow 12 + 2 + 31 + 24 + 2 + 59 + 16 + 199 + 14 + 23$$

$$\Rightarrow 382 \text{ m}$$

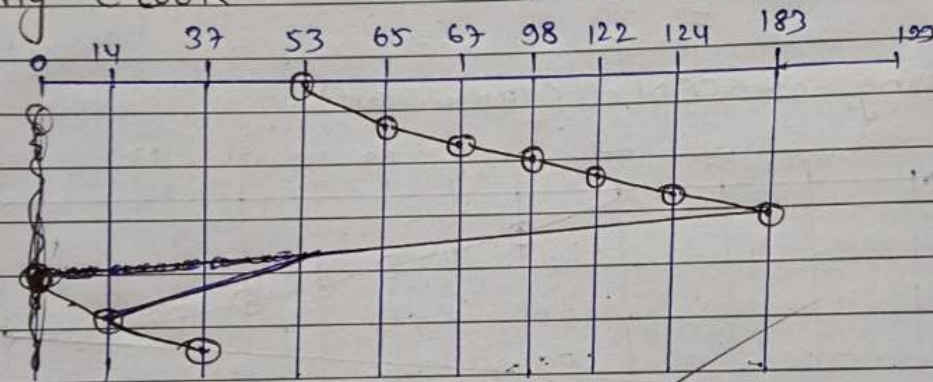
• Using LOOK Algorithm :-



- Total Track movement :

$$\begin{aligned}
 & (65-53) + (67-65) + (98-67) + (122-98) + (124-122) \\
 & + (183-124) + (183-97) + (37-14) \\
 & = 12 + 2 + 31 + 24 + 2 + 59 + 146 + 23 \\
 & = \underline{299 \text{ AS}}
 \end{aligned}$$

- using C-look :



- Total Track movement :

$$\begin{aligned}
 & (65-53) + (67-65) + (98-67) + (122-98) + (124-122) \\
 & + (183-124) + (183-14) + (37-14) \\
 & = 12 + 2 + 31 + 24 + 2 + 59 + 169 + 23 \\
 & = \underline{322}
 \end{aligned}$$

Total Track movement = 322

AS

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